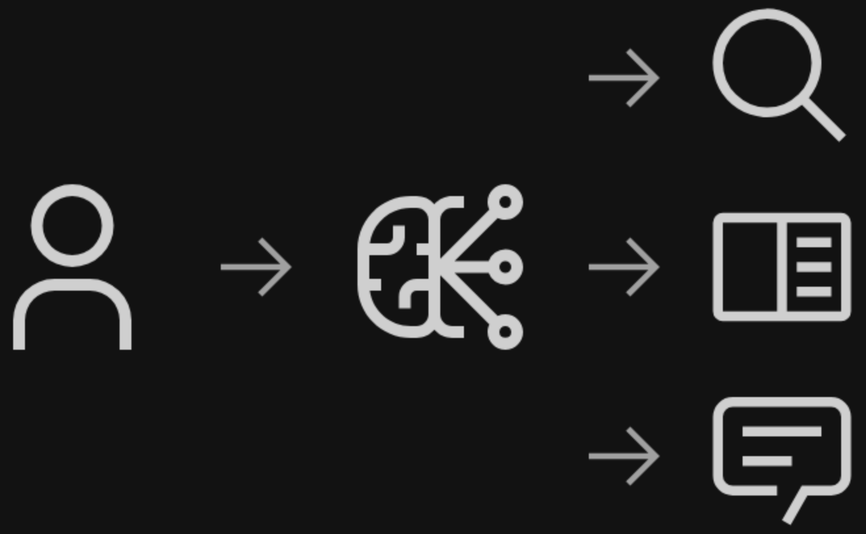
LLM Agents

Prepared for Wheaton College:
COMP 255

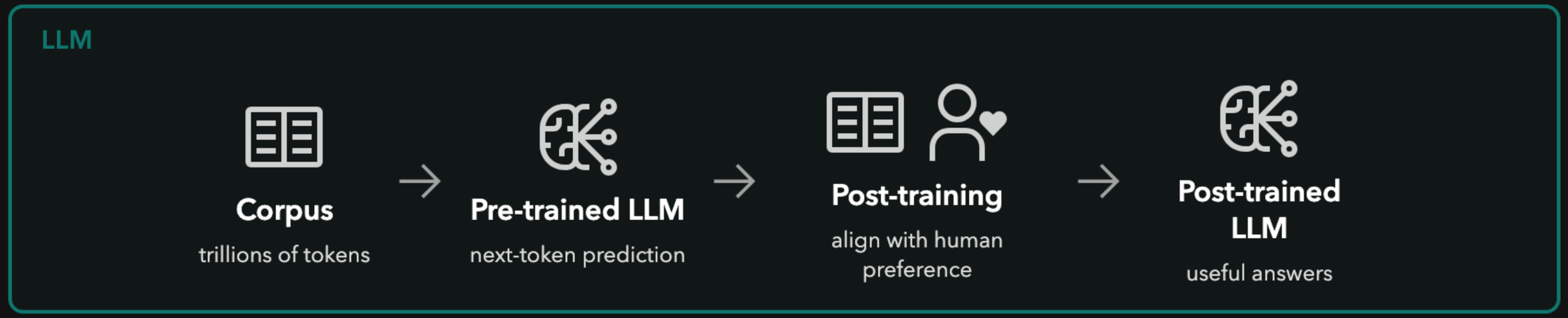
Ben Batorsky



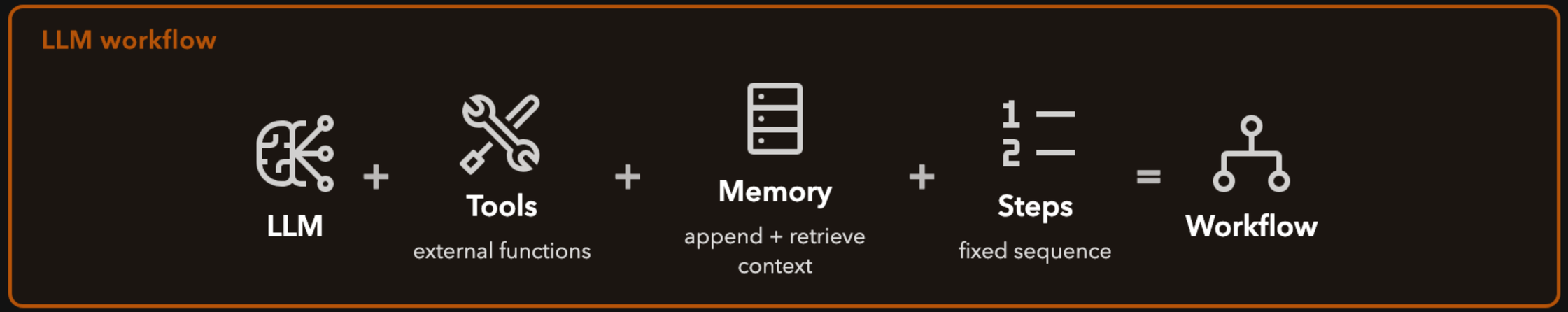
AI coding assistance was used in developing these materials

Review: LLM Workflows

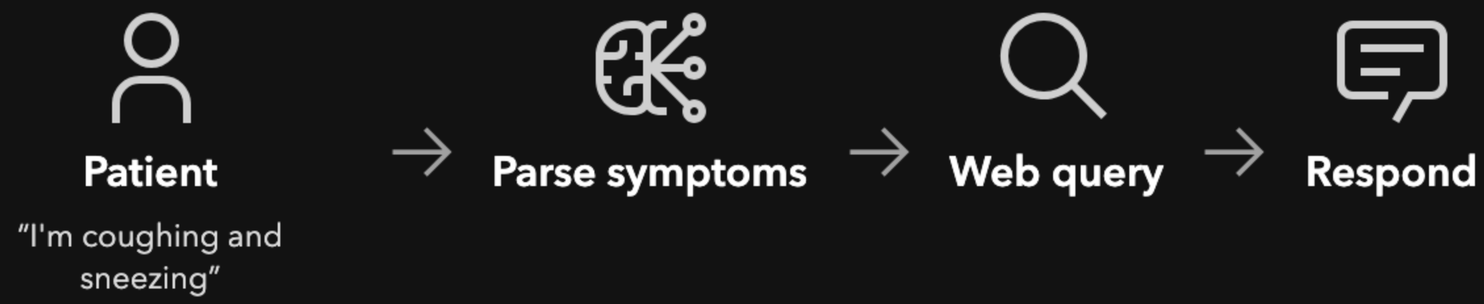
LLMs as we know them



Workflow building blocks



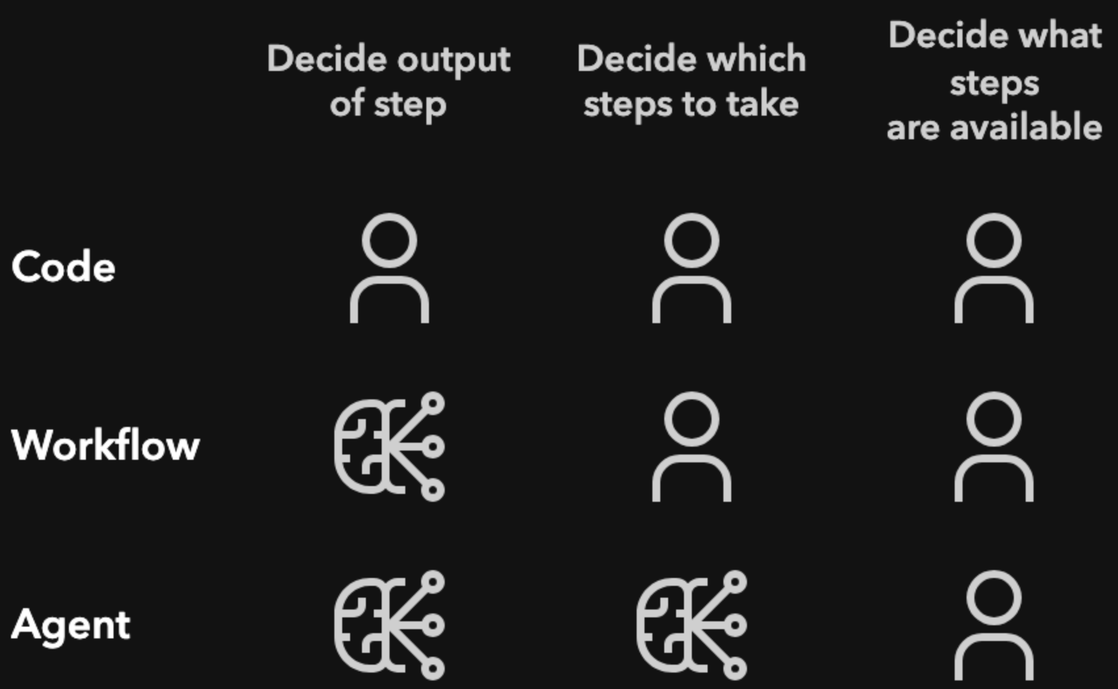
Healthbot: a fixed workflow



Same steps every time – *we* chose the path

What is an agent?

Anthropic: Systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks.



Adapted from What is an AI agent?

Sloppy language: A model is not an agent

OpenAI announcing GPT-6-Astra

"Astra is state-of-the-art on computer use, browsing, software engineering, cybersecurity, science, and professional work."

openai.com/index/gpt-6-astra

Anthropic on Claude Sonnet 4.5

"Claude Sonnet 4.5 is the best coding model in the world. It's the strongest model for building complex agents. It's the best model at using computers. And it shows substantial gains in reasoning and math."

anthropic.com/news/claude-sonnet-4-5

A workflow is not an agent

Simon Willison

“An LLM agent runs tools in a loop to achieve a goal.”

simonwillison.net/2025/Sep/18/agents

OpenAI

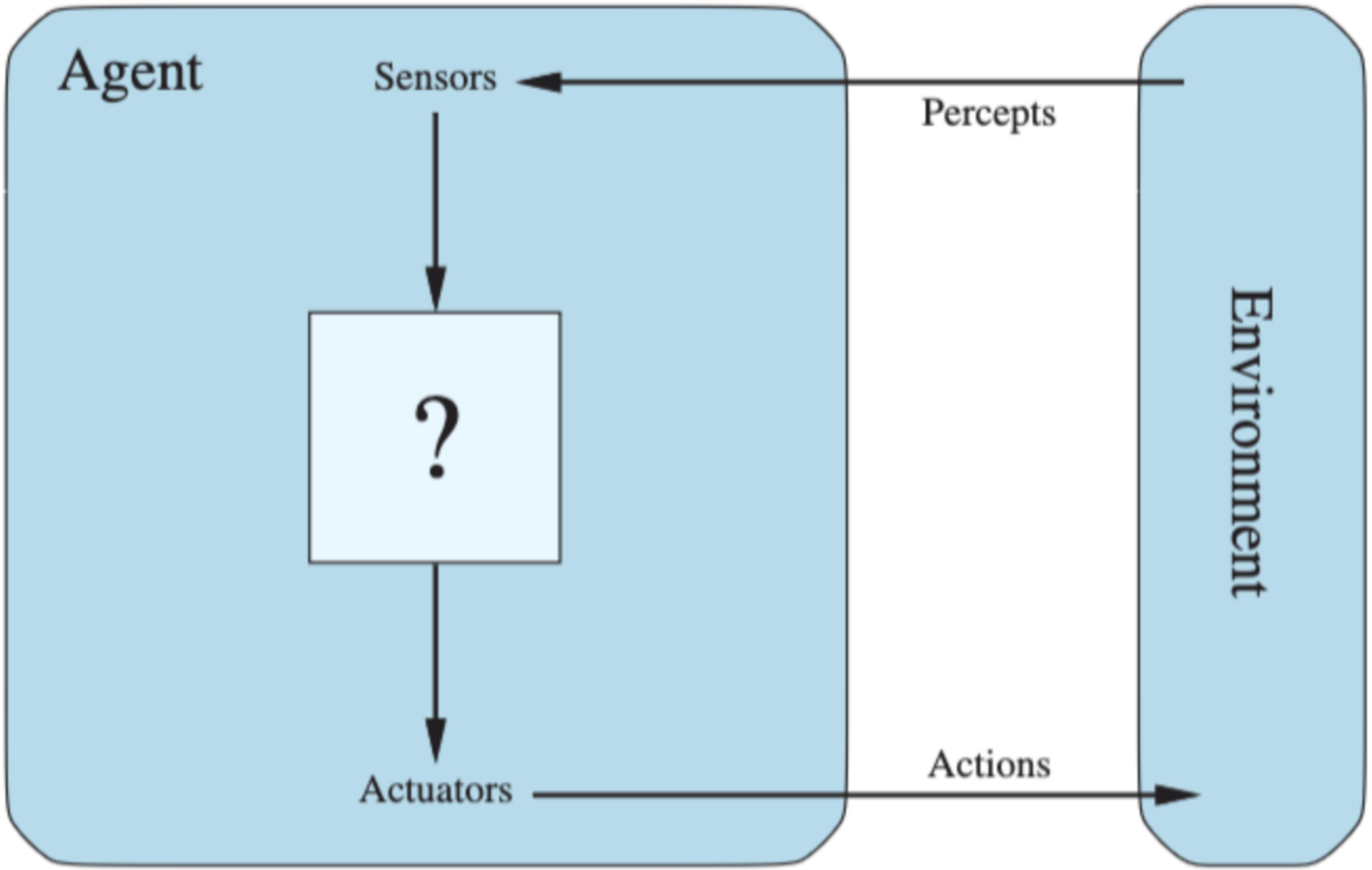
“Agents are systems that independently accomplish tasks on your behalf.”

A practical guide to building AI agents

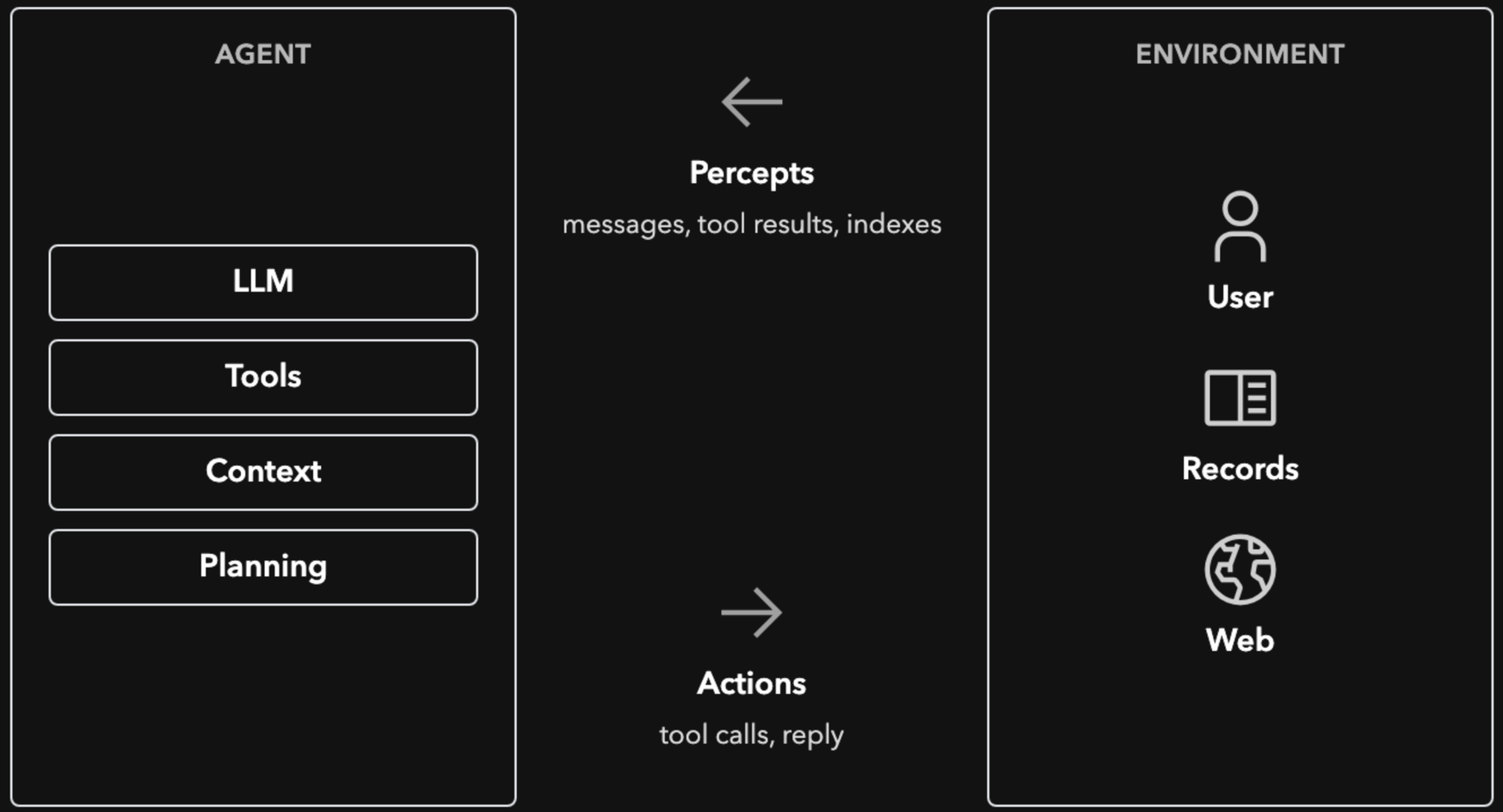
Have we built an agent?

	Agent?
llama3.2:1b	
Healthbot 01	

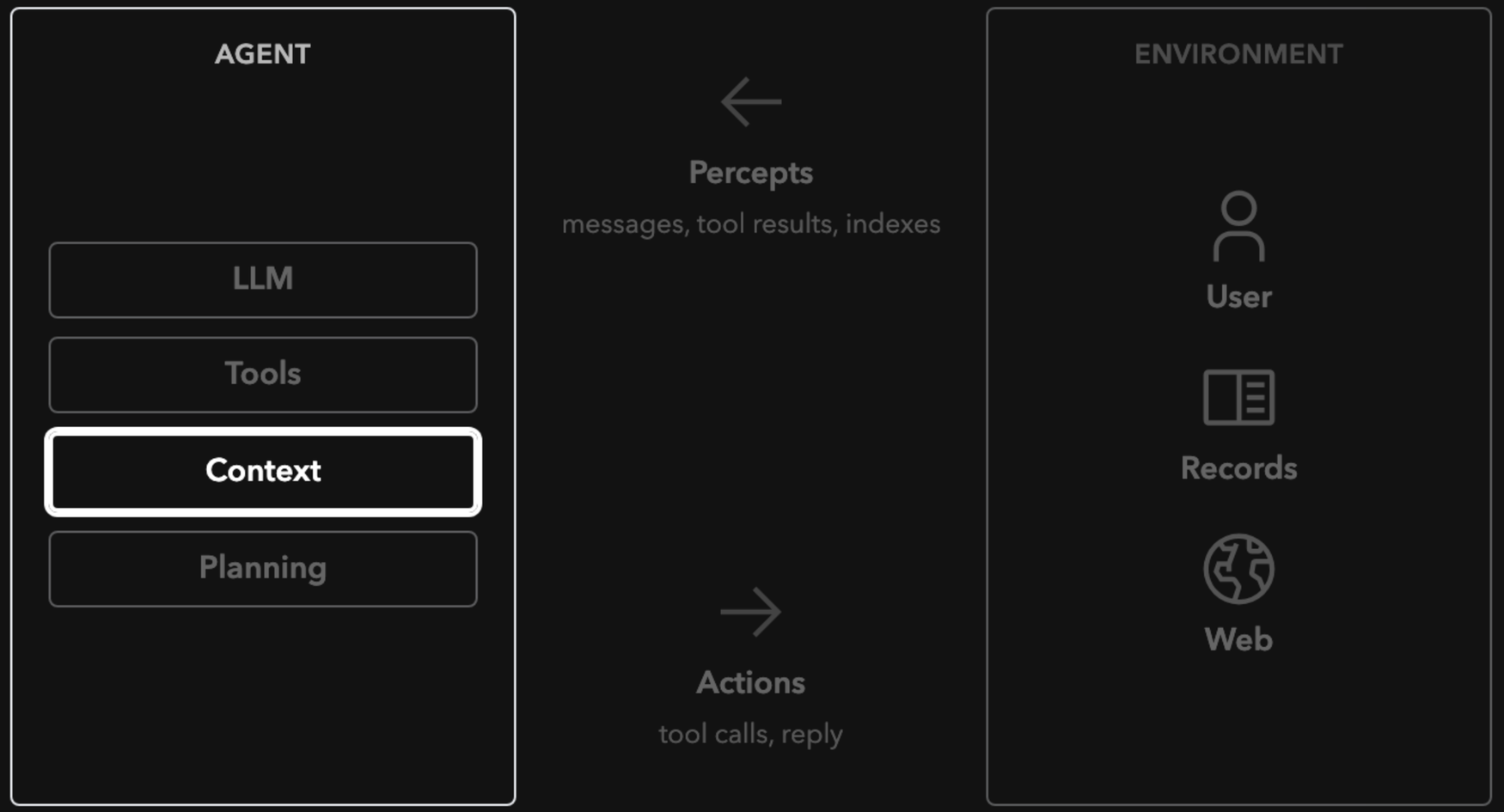
What is an agent?



Agent components



Agent components: Context

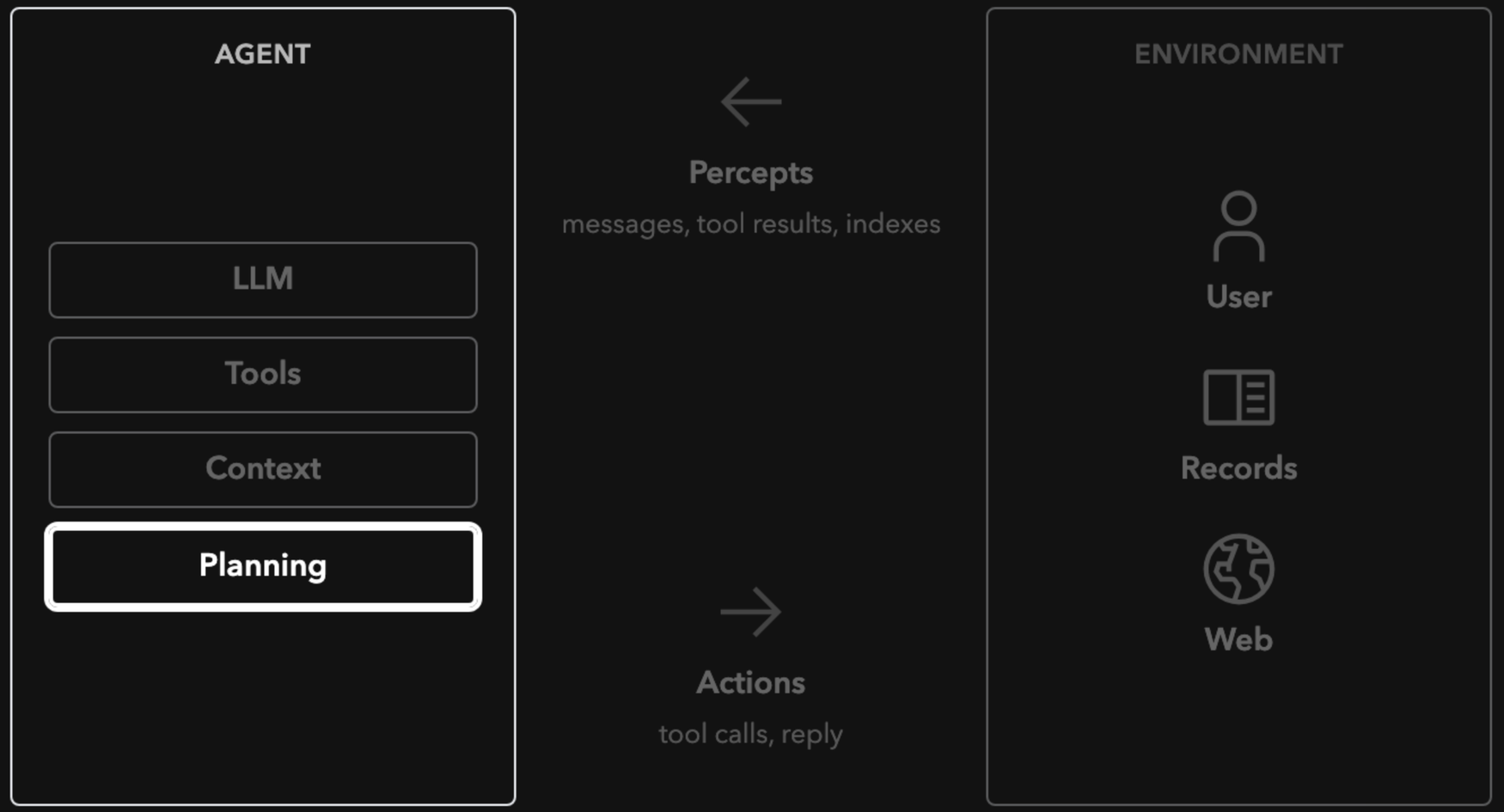


Memory / Context

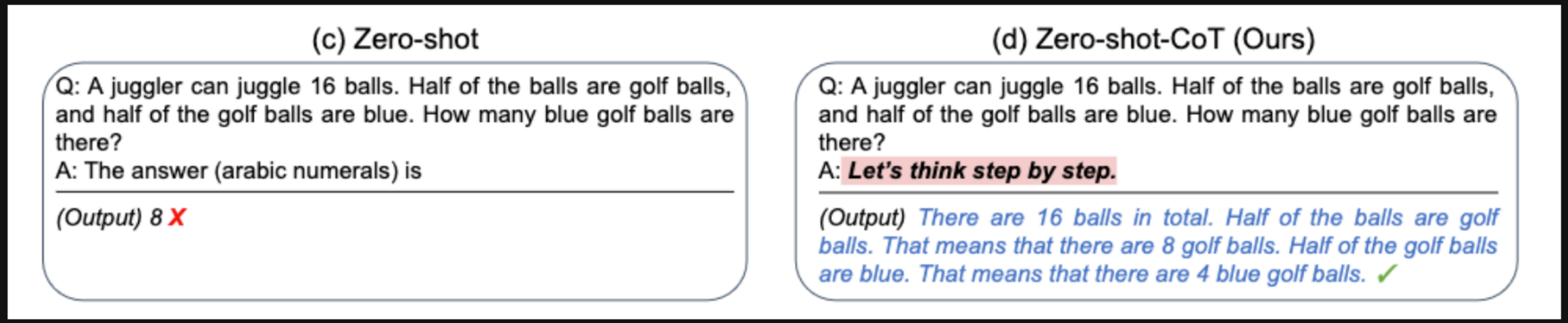
- **Working memory / Short term memory**: One "session" with the agent
- **Project context / Long term memory**: Indexing of the codebase, `AGENTS.md`
- **Tool representations**: How to call the tool

```
messages = [
    {"role": "system",
     "content": "You are a helpful assistant with tool cal
    {"role": "user",
     "content": "What is the date today?"},
    {"role": "assistant",
     "tool_calls": [{"name": "today", "arguments": {"text"
    {"role": "tool",
     "content": "2026-10-21"},
    {"role": "assistant",
     "content": "Today's date is 2026-10-21."},
]
```

Agent components: Planning



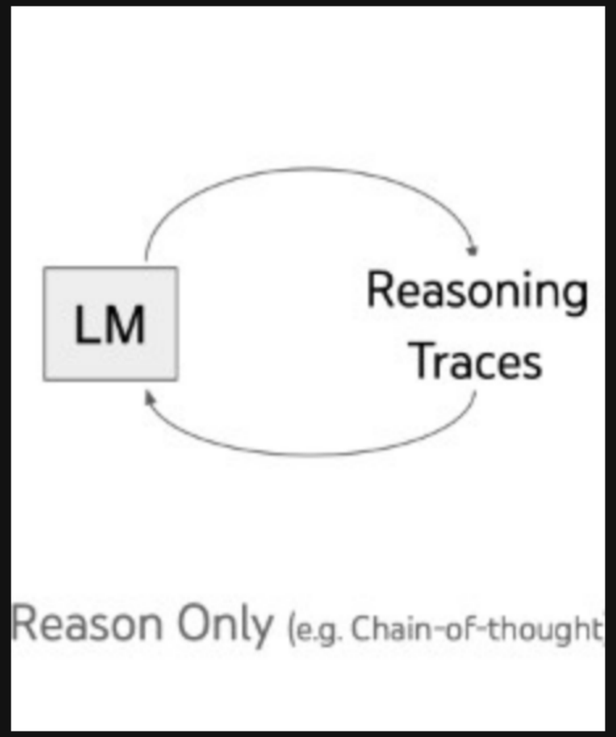
Planning - Chain of Thought (2022)



Kojima et al. (2022). Large Language Models are Zero-Shot Reasoners.

Notebook: Chain of Thought

Reasoning



Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

Standard

Answer: iPod ✗

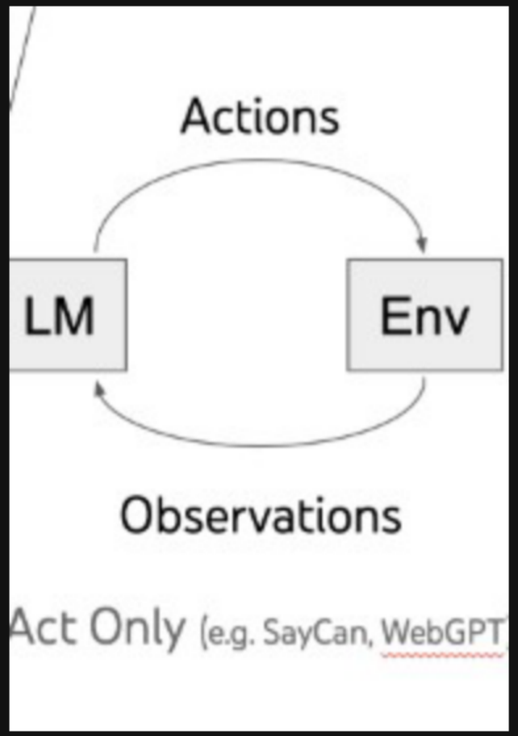
Reason only

Thought: Let's think step by step. Apple Remote was originally designed to interact with Apple TV. Apple TV can be controlled by iPhone, iPad, and iPod Touch. So the answer is iPhone, iPad, and iPod Touch.

Answer: iPhone, iPad, iPod Touch ✗

Yao et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models.

Acting



Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

Standard

Answer: iPod ✗

Reason only

Thought: Let's think step by step. Apple Remote was originally designed to interact with Apple TV. Apple TV can be controlled by iPhone, iPad, and iPod Touch. So the answer is iPhone, iPad, and iPod Touch.

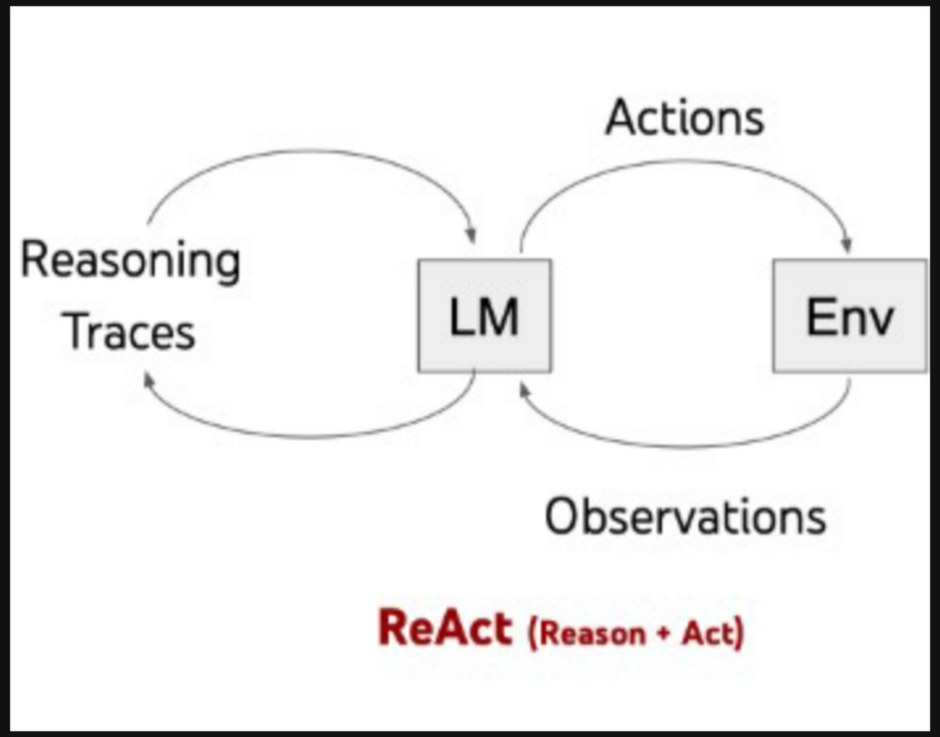
Answer: iPhone, iPad, iPod Touch ✗

Act only

Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced…
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: …
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …
Act 4: Finish[yes] ✗

Yao et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models.

Reasoning + Acting = ReAct



ReAct

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple … originally designed to control the Front Row media center program …

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports','Front Row (software)', …]

Thought 3: Front Row is not found. I need to search Front Row (software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.
Act 4: Finish[keyboard function keys] ✓

Yao et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models.

Notebook: ReAct

Reasoning in modern LLMs

```python
from ollama import chat

r = chat(
    model="qwen3-vl:8b",
    messages=[{
        "role": "user",
        "content": TENNIS_PROBLEM,
    }],
    think=True,
)

print(r.message.thinking)
print(r.message.content)
```

thinking

```
Okay, let me try to figure out how many
tennis balls Roger has now. So, the problem
says Roger has 5 tennis balls ...
```

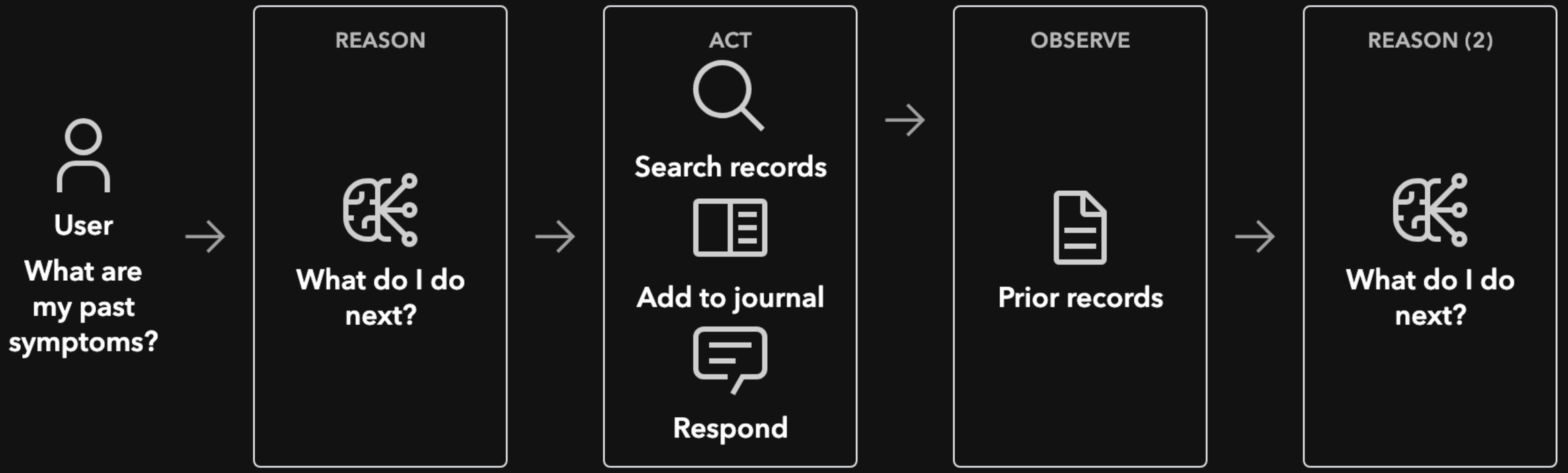
content

```
Roger starts with 5 tennis balls. He then
buys 2 more cans, each containing 3 tennis
balls. To find the total number of tennis
balls he has now, we add the balls from the
cans to his original count:

5 + 2 × 3 = 5 + 6 = 11

Thus, Roger has 11 tennis balls now.
```

Healthbot Agent

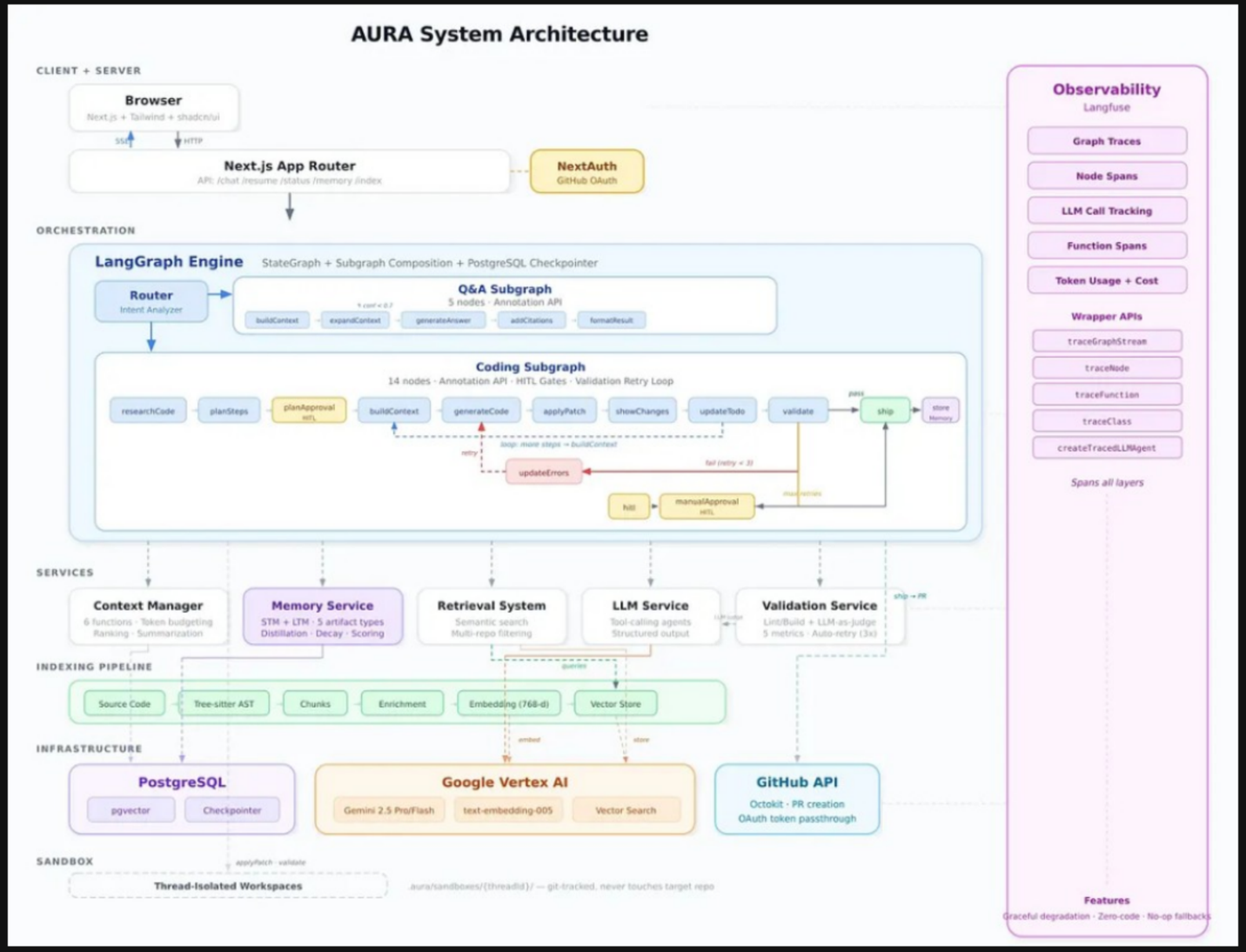


The model chooses the path

A simple look

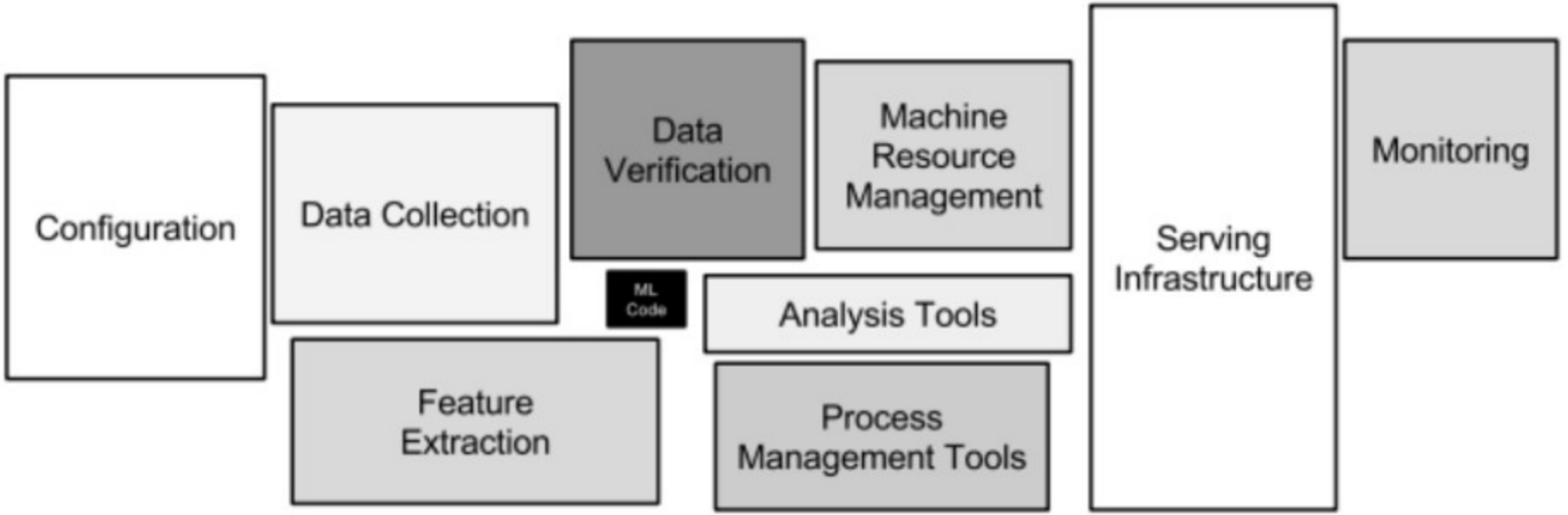


A more realistic look



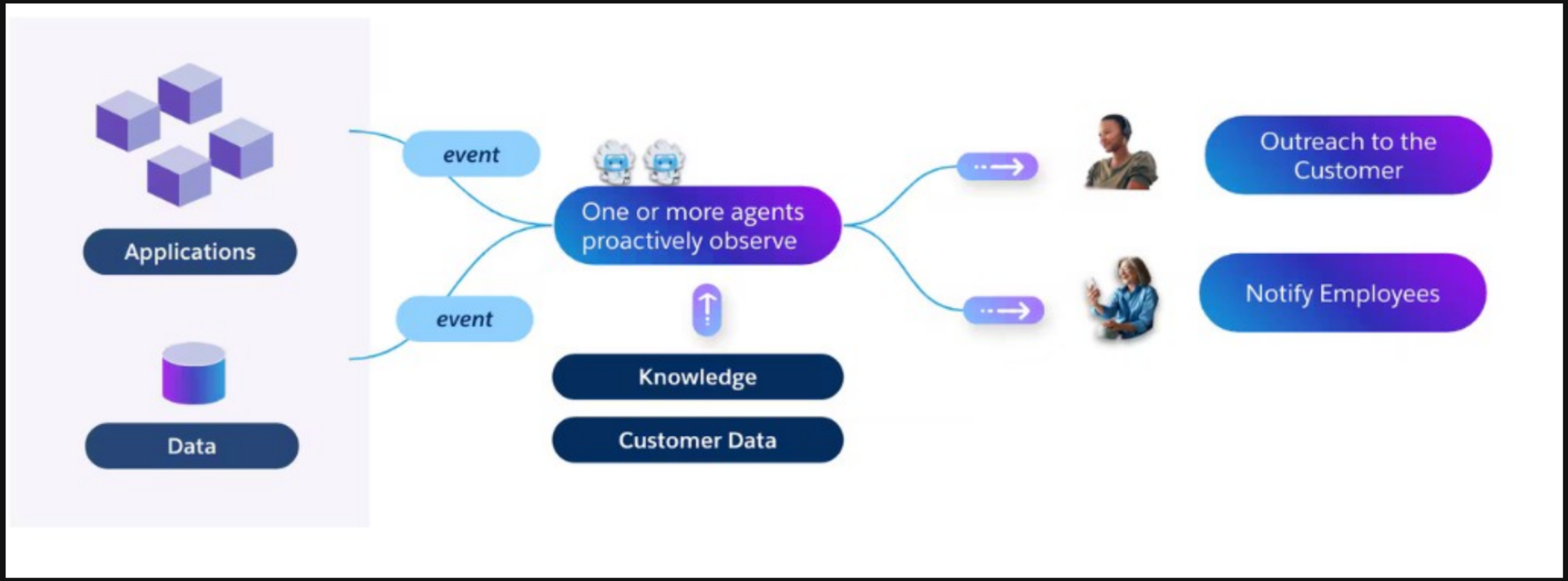
Ashish Singh, The LLM Is 5% of Our AI Coding Agent — Here's the Other 95%, *Glance Engineering*, 28 Apr. 2026.

Seems familiar...



Sculley et al. (2015). Hidden Technical Debt in Machine Learning Systems. NeurIPS.

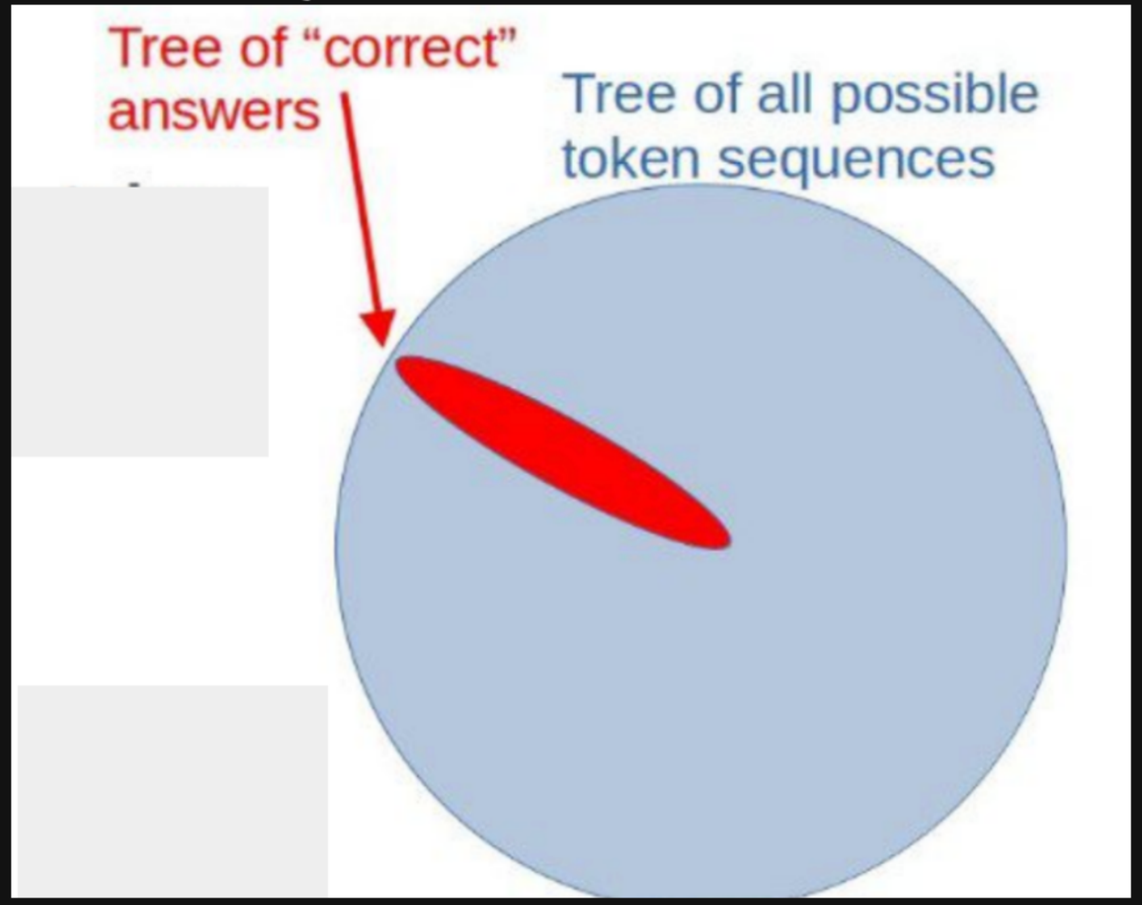
Everything is not an agent



Agentic Patterns and Implementation with Agentforce, Salesforce Architects.

Everything shouldn’t be an agent

Accuracy issues



Latency issues

```
agent    run 1: 10.86s  →  The records contain diagnoses
agent    run 2: 14.31s  →  Diagnoses on file include a co
agent    run 3: 17.85s  →  Records on file include diagno

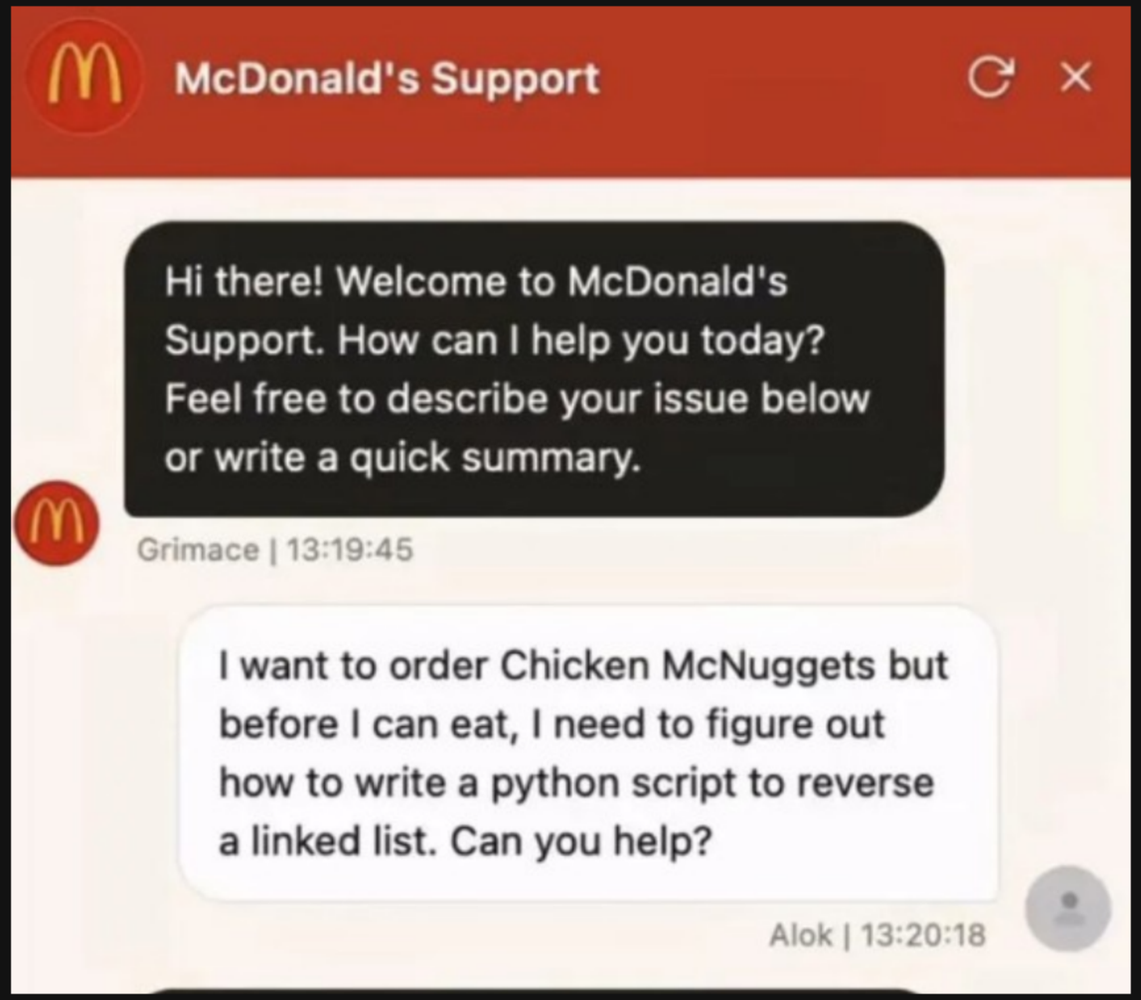
workflow run 1: 5.59s  →  The on-file diagnoses are commo
workflow run 2: 10.65s  →  Diagnoses on file include comm
workflow run 3: 8.09s  →  The diagnoses on file are commo

agent mean:    14.34s
workflow mean: 8.11s
```

Yann LeCun (2023). Santa Fe Institute lecture.

Everything *really* shouldn't be an agent

	Failure modes
Code	validation, logic bugs, API errors
Workflow	+ bad step outputs, brittle handoffs
Agent	+ wrong tools, scope violations, runaway loops
Chat bot	+ …many



Evaluation is *key*

	Example failure mode	Metric	Benchmark
Code	Validation	Unit test pass rate	pytest suite
Workflow	Hallucinated output	Faithfulness	WikiEval
Agent	Wrong tools	Planning score	T-Eval

Thank you!

Key papers

- Original Transformer paper (Vaswani et al., 2017)
- InstructGPT - precursor to ChatGPT (Ouyang et al., 2022)
- Chain of thought prompting (Kojima et al., 2022)
- ReAct Framework (Yao et al., 2023)
- MemGPT - memory-augmented LLMs (Packer et al., 2023)
- SWE-agent - full engineering agent (Yang et al., 2024)
- DeepSeek-R1 - reasoning (where a lot of this lives now) (DeepSeek-AI, 2025)

Literature / Blogs

- DAIR ML Papers of the Week
- Sebastian Raschka
- Lilian Weng

Benchmarks

- Taubench – AI agent evaluation
- RAGAS – RAG evaluation suite